

North East University Bangladesh

Department of Computer Science And Technology

Project Proposal for ERP System

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Introduction:

In today's fast-paced business environment, managing different departments and workflows efficiently is essential for growth and stability. An Enterprise Resource Planning (ERP) system is a centralized platform that integrates various core business processes like human resources, inventory, sales, and branch management into a unified system.

This project, **ERP Management System**, is designed to automate and streamline tasks across multiple departments such as HR, Inventory, Sales, and Branch Operations through defined user roles and permissions. It will help improve communication, coordination, and decision-making inside an organization.

Objectives:

- To develop a centralized ERP system with multi-user access and role-based functionalities.
- To manage products, sales, users, and leaves across multiple branches.
- To reduce manual paperwork and improve the efficiency of internal operations.
- To implement proper access control for different users such as HR, CEO, Employee, Inventory Manager, and Branch Manager.
- To ensure transparency and smooth workflow between departments.

Key Features / Functionalities:

User Roles and Access:

- **CEO**: Overview of all operations, reports, and summaries.
- **HR**: Manage employees, approve/reject leave applications, view attendance and employee data.
- **Employee**: View product stock in their branch, sell products, request leave.
- **Inventory Manager**: Manage product stocks, update inventory, respond to branch product requests.
- **Branch Manager**: Manage users of their branch, request products, view branch-level reports.

Module Descriptions:

1. **User Management:**
    - Role-based login system
    - Secure session management.
    - CRUD operations for employee data by HR or Branch Manager.

2. **Inventory Management:**
    - Add, update, delete products (Inventory Manager).
    - Branch-wise product stock tracking.
    - Product transfer between inventory and branches.

3. Sales Management (Employee Role):

- Sell available products.
- View daily/monthly sales history.
- Branch-based sales filtering.

4. Leave Management (HR Role):

- Leave request by employees.
- Approval/rejection by HR.
- Leave history and tracking.

5. Branch Management:

- Request product stock from Inventory Manager.
- Manage users in the specific branch.
- Monitor branch activities and reports.

Technologies Used:

- **Frontend**: HTML, CSS, Bootstrap
- **Client-side Scripting**: JavaScript
- **Backend**: Core PHP
- **Database**: MySQL
- **Styling Framework**: Bootstrap
- **Server**: Apache (XAMPP or LAMP)

Conclusion:

The ERP Management System will serve as a complete solution for companies to manage their internal departments effectively. With clearly defined user roles and centralized data handling, the project will bring automation, accuracy, and ease of management, making it a reliable system for modern businesses.